

Oksidlanish-qaytarilish reaksiyalari

9

oksidlanish darajasi · elektron balans va koeffitsiyentlar · oksidlovchi/qaytaruvchi · OQR turlari (molekulyararo, ichki molekulyar, disproporsiya) · aktivlik qatori

NIMALARNI TEKSHIRADI

- darajani aniqlash (murakkab tuzlarda ham)
- elektron balans: koeffitsiyentlar yig'indisi
- mol-massa-hajm hisoblari e-balans orqali
- oksidlovchi/qaytaruvchini topish va tasniflash
- parametrli (alkan-elektron, teskari) masalalar

VIZUAL KO'NIKMALAR

- daraja/balans jadvallarini o'qish (A: 17; B: 5, 17, 29)
- $n(e)-n(Me)$ grafigi bilan ishlash (B: 27, 32)
- aktivlik qatori va tajriba rasmlari (A: 32; B: 28)

TEZ-TEZ UCHRAYDIGAN XATOLAR

- koeffitsiyent bilan indeksni adashtirish
- «e bergan — qaytariladi» degan teskari tasavvur
- disproporsiyani molekulyararo OQR bilan chalkashtirish
- HCl kabi moddaning ikki vazifasini (qaytaruvchi + muhit) unutish

Qism	Topshiriqlar	Turi	Vaqt
1	1-32	Y1 — yopiq test (A-D)	100 daqiqa
2	33-35	Y2 — moslashtirish (A-F)	
3	36-40	O1 — qisqa javob	
4	41-43	O2 — yozma ish (har biri 25 ball)	80 daqiqa

1-QISM · YOPIQ TEST (1–32 · to'rt variantdan bittasi to'g'ri)

1. Oksidlanish-qaytarilish reaksiyalari deb qanday reaksiyalarga aytiladi?

- A) issiqlik ajralib chiqadigan reaksiyalarga
- B) cho'kma hosil bo'ladigan reaksiyalarga
- C) faqat kislorod ishtirok etadigan reaksiyalarga
- D) elementlarning oksidlanish darajalari o'zgaradigan reaksiyalarga

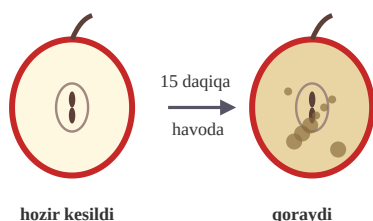
2. Oksidlanish jarayoni deb nimaga aytiladi?

- A) zarrachaning elektron BERISHIGA
- B) zarrachaning elektron OLISHIGA
- C) moddaning kislorod bilan birikishiga
- D) moddaning parchalanishiga

3. H_2SO_4 birikmasidagi oltingugurtning oksidlanish darajasini aniqlang.

- A) +4 B) +6 C) –2 D) +2

4. Rasmga qarang: kesilgan olma bir necha daqiqada qorayadi. Buning kimyoviy sababi nimada?



- A) olma suvi bug'lanib ketadi
- B) olma tarkibidagi moddalar havo kislorodi ta'sirida oksidlanadi
- C) olma tarkibida temir zanglaydi
- D) yorug'lik olmani kuydiradi

5. Reaksiyada QAYTARUVCHI vazifasini bajargan zarracha ...

- A) elektron oladi va o'zi qaytariladi
- B) elektron beradi va o'zi qaytariladi
- C) elektron beradi va o'zi oksidlanadi
- D) elektron olmaydi ham, bermaydi ham

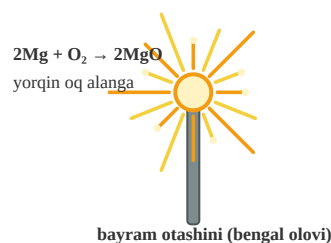
6. KMnO_4 birikmasidagi marganesning oksidlanish darajasini aniqlang.

- A) +4 B) +7 C) +2 D) +6

7. $\text{Zn}^0 \rightarrow \text{Zn}^{2+}$ o'zgarishi qanday jarayon?

- A) oksidlanish — 2 ta elektron berildi
- B) qaytarilish — 2 ta elektron olindi
- C) almashinish
- D) dissotsiatsiya

8. Rasmda bayram otashini (bengal olovi) ko'rsatilgan: magniy kukuni yorqin oq alanga bilan yonadi ($2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$). Bu reaksiyada magniy qanday vazifani bajaradi?



- A) oksidlovchi — elektron oladi
- B) qaytaruvchi — elektron berib oksidlanadi
- C) katalizator — o'zi o'zgarmaydi
- D) muhit hosil qiluvchi

9. $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$ reaksiyasida OKSIDLOVCHI qaysi modda?

- A) H_2 B) Cu C) CuO D) H_2O

10. $\text{Al}^0 \rightarrow \text{Al}^{3+}$ o'zgarishida bitta alyuminiy atomi nechta elektron beradi?

- A) 1 B) 2 C) 13 D) 3

11. Quyidagi o'zgarishlarning qaysilari OKSIDLANISH hisoblanadi?

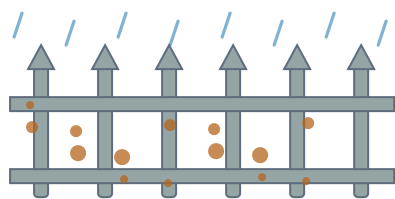
1) $\text{S}^0 \rightarrow \text{S}^{+4}$; 2) $\text{Cu}^{+2} \rightarrow \text{Cu}^0$; 3) $\text{N}^{-3} \rightarrow \text{N}^{+2}$.

- A) faqat 1
- B) 2 va 3
- C) 1 va 3
- D) 1 va 2

12. Ammiak (NH_3) tarkibidagi azotning oksidlanish darajasi qanday?

- A) +3 B) 0 C) +5 D) –3

13. Rasmda zanglagan temir panjara ko'rsatilgan. Zanglash (korroziya) jarayonida temir qanday vazifani bajaradi?



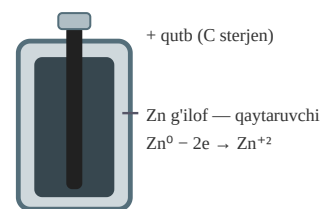
yomg'irda zanglagan panjara

- A)** oksidlovchi — elektron oladi
B) qaytaruvchi — elektron berib Fe^{+3} gacha oksidlanadi
C) katalizator
D) hech qanday — zanglash fizik jarayon
- 14.** $\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$ reaksiyasi tenglashtirilganda barcha koeffitsiyentlar yig'indisini toping.
A) 3 **B)** 4 **C)** 5 **D)** 7
- 15.** 0,3 mol rux to'liq oksidlanib Zn^{2+} ga o'tganda necha mol elektron beradi?
A) 0,3 **B)** 1,2 **C)** 0,15 **D)** 0,6
- 16.** Quyidagi reaksiyalardan qaysi biri DISPROPORSIYAGA (bir element ham oksidlanib, ham qaytarilishiga) misol?
A) $\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{HCl} + \text{HClO}$
B) $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$
C) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
D) $\text{Fe} + \text{S} \rightarrow \text{FeS}$
- 17.** Jadvalda reaksiya ishtirokchilarining darajalari berilgan:

Element	boshlang'ich	oxirgi
Mn	+7	+2
Fe	+2	+3

OKSIDLOVCHI qaysi element?

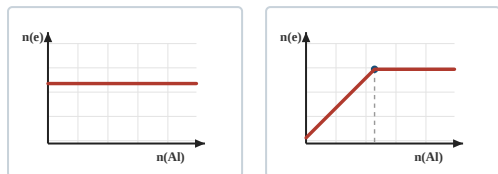
- A)** Fe — darajasi ortgan
B) ikkalasi ham
C) Mn — darajasi pasaygan
D) hech qaysi
- 18.** Rasmda cho'ntak batareykasi ko'rsatilgan: unda rux g'ilof asta-sekin yemiriladi va tok hosil bo'ladi. Rux qanday vazifani bajaradi?



cho'ntak batareykasi (kesim)

- A)** oksidlovchi — elektron yig'adi
B) izolyator
C) katalizator
D) qaytaruvchi — elektron berib, tashqi zanjirga tok beradi
- 19.** $\text{Fe} + \text{Cl}_2 \rightarrow \text{FeCl}_3$ reaksiyasi tenglashtirilganda barcha koeffitsiyentlar yig'indisini toping.
A) 7 **B)** 5 **C)** 6 **D)** 4
- 20.** Erkin holdagi oddiy moddalarda (Cl_2 , O_2 , Fe) elementlarning oksidlanish darajasi qanday?
A) 0 **B)** +1 **C)** -1 **D)** guruh raqamiga teng
- 21.** $\text{Al} + \text{O}_2 \rightarrow \text{Al}_2\text{O}_3$ reaksiyasi elektron balans bilan tenglashtirilganda QAYTARUVCHI oldida qanday koeffitsiyent turadi?
A) 2 **B)** 4 **C)** 3 **D)** 1
- 22.** Kislotali muhitda 0,1 mol KMnO_4 ($\text{Mn}^{+7} \rightarrow \text{Mn}^{+2}$) qaytarilganda necha mol elektron oladi?
A) 0,1 **B)** 0,5 **C)** 0,7 **D)** 0,2
- 23.** Quyidagi o'zgarishlardan qaysi biri QAYTARILISH jarayoni?
A) $\text{S}^0 \rightarrow \text{S}^{+4}$
B) $\text{Fe}^{+2} \rightarrow \text{Fe}^{+3}$
C) $\text{Cu}^{2+} \rightarrow \text{Cu}^0$
D) $\text{Cl}^- \rightarrow \text{Cl}_2$
- 24.** $\text{Cu} + 2\text{AgNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + 2\text{Ag}$. 0,1 mol mis reaksiyaga to'liq kirishganda necha gramm kumush ajralib chiqadi? ($M(\text{Ag})=108$)
A) 10,8 **B)** 108 **C)** 5,4 **D)** 21,6
- 25.** Quyidagi moddalardan qaysi birida element ENG YUQORI darajada bo'lib, u faqat OKSIDLOVCHI bo'la oladi?
A) MnO_2 (Mn^{+4})
B) MnSO_4 (Mn^{+2})
C) Mn (Mn^0)
D) KMnO_4 (Mn^{+7})

26. Alyuminiy mol soni ortib borganda u beradigan elektron mollari qanday o'zgaradi? To'g'ri grafikni tanlang.



A)

B)

C)

D)

27. 5,4 g alyuminiy to'liq oksidlanganda necha mol elektron beradi? ($M(\text{Al})=27$)

A) 0,2 B) 1,8 C) 0,4 D) 0,6

28. $2\text{KClO}_3 \rightarrow(t) 2\text{KCl} + 3\text{O}_2$ reaksiyasi qaysi turdagi OQR?

A) ichki molekulyar
B) disproporsiya
C) molekulyararo
D) OQR emas

29. $\text{H}_2\text{S} + \text{O}_2 \rightarrow \text{SO}_2 + \text{H}_2\text{O}$ reaksiyasi tenglashtirilganda barcha koeffitsiyentlar yig'indisini toping.

A) 9 B) 7 C) 5 D) 11

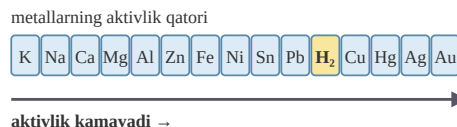
30. Temir buyumni korroziyadan (zanglashdan) himoya qilishning qaysi usuli OKSIDLOVCHI bilan aloqani uzishga asoslangan?

A) sirtini bo'yoq bilan qoplash
B) buyumni tez-tez yuvib turish
C) buyumni qizdirish
D) kislota bilan artish

31. 8 g mis (II) oksidi vodorod bilan to'liq qaytarildi: $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$. Necha gramm mis hosil bo'ladi? ($M(\text{CuO})=80$, $M(\text{Cu})=64$)

A) 8 B) 6,4 C) 3,2 D) 12,8

32. Rasmda metallarning aktivlik qatori berilgan. Undan foydalanib aniqlang: qaysi metall CuSO_4 eritmasidan misni siqib chiqara OLADI?



A) Ag B) Au C) Fe D) Hg

2-QISM · GURUHLANGAN SAVOL (33–35 · A–F ro'yxatidan tanlang)

Bayram otashinida 2,4 g magniy kukuni to'liq yondi: $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$. ($M(\text{Mg})=24$, $M(\text{MgO})=40$.) 33–35-savollarga A–F ro'yxatidan javob tanlang.

33. Magniy bergan elektronlar necha mol?

34. Hosil bo'lgan magniy oksidining massasi (g) qancha?

35. Sarflangan kislorodning hajmi (l, n.sh.) qancha?

Javoblar ro'yxati: A) 4 · B) 0,2 · C) 2,24 · D) 1,12 · E) 0,1 · F) 2

3-QISM · QISQA JAVOBLI SAVOLLAR (36–40)

36. Erkin azot (N_2) molekulasidagi azotning oksidlanish darajasini yozing. Javob:

37. $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+}$ o'zgarishida temir ioni nechta elektron beradi? Javob:

38. $\text{Al} + \text{S} \rightarrow \text{Al}_2\text{S}_3$ reaksiyasi tenglashtirilganda barcha koeffitsiyentlar yig'indisini toping. Javob:

39. 0,2 mol magniy to'liq oksidlanganda beradigan elektron mollarini toping. Javob:

40. 3,2 g mis AgNO_3 ning mo'l eritmasiga tushirildi. Ajralib chiqqan kumushning massasini (g) toping. ($M(\text{Cu})=64$, $M(\text{Ag})=108$) Javob:

4-QISM · YOZMA ISH (41–43 · har biri 25 ball)**41-topshiriq**

6,5 g rux mis (II) sulfatning mo'l eritmasiga tushirildi: $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$. Bandlar ketma-ket yechiladi — har biri keyingisiga asos bo'ladi. ($M(\text{Zn})=65$, $M(\text{Cu})=64$)

- Reaksiyaning elektron balansini yozing: kim e beradi, kim oladi? (4 ball · M3+A1)
- Ruxning miqdorini (mol) toping. (5 ball · M3+A2)
- Ajralib chiqqan misning massasini (g) hisoblang. (7 ball · M4+A3)
- Jarayonda necha mol elektron almashindi? (4 ball · M2+A2)
- Nega bu reaksiya OQR hisoblanadi? Bir jumla bilan izohlang. (5 ball · M3+A2)

42-topshiriq

Qishloq hovlisidagi temir panjara yomg'irli mavsumda tez zanglaydi. Quyidagi savollarga MULOHAZA yuritib javob yozing (hisob talab qilinmaydi).

- Zanglash jarayonining OQR mohiyatini yozing: qaytaruvchi va oksidlovchini ko'rsatib, umumiy tenglamani keltiring. (13 ball · M13)
- Nega namlik (yomg'ir) zanglashni keskin tezlashtiradi? (9 ball · M9)
- Panjarani himoya qilishning bitta usulini tavsiya qiling va sababini ayting. (3 ball · M3)

43-topshiriq

O'quvchi uchta stakanda tajriba o'tkazdi; kuzatuvlar jadvalda:

Stakan	Metall	Eritma	Kuzatuv
1	Cu	FeSO_4	o'zgarish yo'q
2	Fe	CuSO_4	qizil qatlam
3	Zn	CuSO_4	qizil qatlam

Bandlar ketma-ket yechiladi. ($M(\text{Fe})=56$, $M(\text{Cu})=64$)

- Nega 1-stakanda reaksiya bormadi? Aktivlik qatori orqali izohlang. (4 ball · M3+A1)
- 2-stakan uchun tenglama yozing va 11,2 g temirdan ajraladigan mis massasini toping. (9 ball · M5+A4)
- 2-stakan jarayonida almashingan elektron mollarini toping. (6 ball · M3+A3)
- Uchala kuzatuvga tayanib Zn, Fe, Cu ni aktivlik tartibida joylashtiring. (6 ball · M4+A2)

A-variant · Javoblar kaliti va yechimlar

№	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Javob	D	A	B	B	C	B	A	B	C	D	C	D	B	C	D	A

№	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
Javob	C	D	A	A	B	B	C	D	D	C	D	A	A	A	B	C

33-35: 33 → B, 34 → A, 35 → D · **36-40:** 36 → 0, 37 → 1, 38 → 6, 39 → 0,4, 40 → 10,8

1. (D) OQR belgisi — kamida ikkita elementning oksidlanish darajasi o'zgarishi (elektron almashinishi).

2. (A) Oksidlanish — e berish (daraja ortadi); qaytarilish — e olish (daraja pasayadi).

3. (B) $2(+1) + x + 4(-2) = 0 \rightarrow x = +6$.

4. (B) Kislorod — oksidlovchi: olma to'qimasidagi moddalarni oksidlab, qoramtir mahsulotlar hosil qiladi. Limon sharbati (vitamin C — qaytaruvchi) bu jarayonni sekinlashtiradi.

5. (C) Qaytaruvchi e beradi (o'zi oksidlanadi) va sherigini qaytaradi.

6. (B) $+1 + x + 4(-2) = 0 \rightarrow x = +7$. Shu sababli KMnO_4 — kuchli oksidlovchi.

7. (A) Daraja 0 dan +2 ga ortdi → Zn 2e berdi → oksidlanish.

8. (B) $\text{Mg}^0 - 2e \rightarrow \text{Mg}^{+2}$ (oksidlandi, ya'ni qaytaruvchi); $\text{O}_2 + 4e \rightarrow 2\text{O}^{2-}$ (oksidlovchi). Yorug'lik — reaksiya energiyasi.

9. (C) $\text{Cu}^{+2} + 2e \rightarrow \text{Cu}^0$: mis oksidi e oldi — oksidlovchi. H_2 — qaytaruvchi.

10. (D) Daraja 0 dan +3 ga ortdi → 3 e berildi.

11. (C) Oksidlanish — daraja ortishi: 1 ($0 \rightarrow +4$) va 3 ($-3 \rightarrow +2$). 2 — qaytarilish.

12. (D) $x + 3(+1) = 0 \rightarrow x = -3$.

13. (B) $4\text{Fe} + 3\text{O}_2 + 6\text{H}_2\text{O} \rightarrow 4\text{Fe}(\text{OH})_3$: temir e beradi (qaytaruvchi), kislorod oladi (oksidlovchi).

14. (C) $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ ($\text{Mg} - 2e \times 2$; $\text{O}_2 + 4e \times 1$). Jami: $2+1+2 = 5$.

15. (D) $n(e) = 0,3 \cdot 2 = 0,6 \text{ mol}$.

16. (A) Cl^0 bir vaqtda -1 (HCl) va $+1$ (HClO) ga o'tadi — disproporsiya.

17. (C) Mn: $+7 \rightarrow +2$ (5 e oldi) — oksidlovchi. Fe: $+2 \rightarrow +3$ (e berdi) — qaytaruvchi.

18. (D) Batareyka — OQRdan tok oladi: $\text{Zn}^0 - 2e \rightarrow \text{Zn}^{+2}$ (qaytaruvchi); elektronlar tashqi zanjir orqali oqadi.

19. (A) $2\text{Fe} + 3\text{Cl}_2 \rightarrow 2\text{FeCl}_3$ ($\text{Fe} - 3e \times 2$; $\text{Cl}_2 + 2e \times 3$). Jami: $2+3+2 = 7$.

20. (A) Oddiy moddada element «o'z-o'zi» bilan bog'langan — daraja 0.

21. (B) $\text{Al} - 3e \times 4$; $\text{O}_2 + 4e \times 3 \rightarrow 4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$. Qaytaruvchi (Al) oldida 4.

22. (B) Har bir Mn^{+7} 5 e oladi → $0,1 \cdot 5 = 0,5 \text{ mol e}$.

23. (C) Qaytarilish — e olish, daraja pasayishi: $+2 \rightarrow 0$.

24. (D) $\text{Ag} = 0,1 \cdot 2 = 0,2 \text{ mol} \rightarrow 0,2 \cdot 108 = 21,6 \text{ g}$.

25. (D) Eng yuqori darajadagi element boshqa e ololmaydi berishga esa «yuqoriroq joy» yo'q — faqat oksidlovchi.

26. (C) $n(e) = 3 \cdot n(\text{Al})$ — noldan chiquvchi to'g'ri chiziq.

27. (D) $n(\text{Al}) = 0,2 \text{ mol} \rightarrow e = 0,2 \cdot 3 = 0,6 \text{ mol}$.

28. (A) Oksidlovchi ham, qaytaruvchi ham KClO_3 ning ichida (Cl^{+5} va O^{2-}) — ichki molekulyar OQR.

29. (A) $2\text{H}_2\text{S} + 3\text{O}_2 \rightarrow 2\text{SO}_2 + 2\text{H}_2\text{O}$ ($\text{S}^{2-} - 6e \times 2$; $\text{O}_2 + 4e \times 3$). Jami: $2+3+2+2 = 9$.

30. (A) Bo'yoq qatlami havo kislorodi va namlikni (oksidlovchilarni) metallardan ajratib turadi.

31. (B) $n(\text{CuO}) = 0,1 \text{ mol} \rightarrow \text{Cu} = 0,1 \cdot 64 = 6,4 \text{ g}$.

32. (C) Qatorda misdan CHAP tomondagi (aktivroq) metallgina Cu^{2+} ni qaytaradi: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$.

33-35. $n(\text{Mg}) = 0,1 \text{ mol}$. 33: $e = 0,1 \cdot 2 = 0,2 \text{ mol (B)}$. 34: $\text{MgO} = 0,1 \cdot 40 = 4 \text{ g (A)}$. 35: $\text{O}_2 = 0,05 \text{ mol} \rightarrow 1,12 \text{ l (D)}$.

36. Oddiy modda — daraja 0.

37. Daraja +2 dan +3 ga ortdi → 1 e.

38. $2\text{Al} + 3\text{S} \rightarrow \text{Al}_2\text{S}_3$ ($\text{Al} - 3e \times 2$; $\text{S} + 2e \times 3$). Jami: $2+3+1 = 6$.

39. $0,2 \cdot 2 = 0,4 \text{ mol e}$.

40. $n(\text{Cu}) = 0,05 \text{ mol} \rightarrow \text{Ag} = 0,1 \text{ mol} \rightarrow 10,8 \text{ g}$.

41-topshiriq. a) Reaksiyaning elektron balansini yozing: kim e beradi, kim oladi? $\text{Zn}^0 - 2e \rightarrow \text{Zn}^{+2}$ (qaytaruvchi); $\text{Cu}^{+2} + 2e \rightarrow \text{Cu}^0$ (oksidlovchi). **b)** Ruxning miqdorini (mol) toping. $n(\text{Zn}) = 6,5/65 = 0,1 \text{ mol}$. **c)** Ajralib chiqqan misning massasini (g) hisoblang. $\text{Cu} = 0,1 \text{ mol} \rightarrow 6,4 \text{ g}$. **d)** Jarayonda necha mol elektron almashindi? $e = 0,1 \cdot 2 = 0,2 \text{ mol}$. **e)** Nega bu reaksiya OQR hisoblanadi? Bir jumla bilan izohlang. Zn va Cu ning oksidlanish darajalari o'zgardi ($0 \leftrightarrow +2$) — elektron almashindi..

42-topshiriq. a) Zanglash jarayonining OQR mohiyatini yozing: qaytaruvchi va oksidlovchini ko'rsatib, umumiy tenglamani keltiring. $\text{Fe} \rightarrow \text{Fe} - 3\text{e} \rightarrow \text{Fe}^{+3}$, O_2 — oksidlovchi (nam ishtirokida); $4\text{Fe} + 3\text{O}_2 + 6\text{H}_2\text{O} \rightarrow 4\text{Fe}(\text{OH})_3$ (keyin quriydi — zang).. **b)** Nega namlik (yomg'ir) zanglashni keskin tezlashtiradi? Suv elektrolit muhit yaratadi — elektron o'tishi (mikrogalvanik juftlar) osonlashadi;; quruq havoda jarayon juda sekin.. **c)** Panjarani himoya qilishning bitta usulini tavsiya qiling va sababini ayting. Bo'yash (yoki ruxlash): oksidlovchi (O_2 , H_2O) bilan aloqa uziladi..

43-topshiriq. a) Nega 1-stakanda reaksiya bormadi? Aktivlik qatori orqali izohlang. Cu aktivlik qatorida Fe dan keyin — Fe^{+2} ni qaytara olmaydi.. **b)** 2-stakan uchun tenglama yozing va 11,2 g temirdan ajraladigan mis massasini toping. $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$; $n(\text{Fe}) = 0,2 \text{ mol} \rightarrow \text{Cu} = 12,8 \text{ g}$. **c)** 2-stakan jarayonida almashingan elektron mollarini toping. $e = 0,2 \cdot 2 = 0,4 \text{ mol}$. **d)** Uchala kuzatuvga tayanib Zn, Fe, Cu ni aktivlik tartibida joylashtiring. $\text{Zn} > \text{Fe} > \text{Cu}$ (Zn va Fe misni siqib chiqardi; Cu temirni chiqara olmadi)..

Manba: MS spetsifikatsiyasi I.9; darslik (8-9-sinf) OQR bo'limlari — savollar yangi tuzilgan, hayotiy sahnalar (olma, otashin, zang, batareyka) bilan

B-VARIANT · HAQIQIY MS MUHITI ★★★ — imtihon darajasi: ko'p

bosqichli hisoblar va tuzoqlar

1-QISM · YOPIQ TEST (1–32 · to'rt variantdan bittasi to'g'ri)

1. $\text{H}_2\text{S} + \text{H}_2\text{SO}_4 + \text{K}_2\text{Cr}_2\text{O}_7 \rightarrow \text{Cr}_2(\text{SO}_4)_3 + \text{S} + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$ reaksiyasi tenglashtirilganda O'NG tomondagi koeffitsiyentlar yig'indisini aniqlang.

A) 12 B) 8 C) 17 D) 7

2. $\text{KMnO}_4 + \text{HCl}(\text{kons}) \rightarrow \text{KCl} + \text{MnCl}_2 + \text{Cl}_2 + \text{H}_2\text{O}$ reaksiyasi tenglashtirilganda BARCHA koeffitsiyentlar yig'indisini toping.

A) 18 B) 35 C) 17 D) 33

3. $\text{K}_2\text{Cr}_2\text{O}_7$ birikmasida xromning oksidlanish darajasini aniqlang.

A) +3 B) +7 C) +2 D) +6

4. Quyidagi jarayonlarning qaysilari oksidlanish-qaytarilish reaksiyalariga mansub?

1) elektroliz; 2) gidroliz; 3) yonish; 4) neytrallanish; 5) korroziya.

A) 1, 3 va 5
B) 1, 2 va 4
C) 2, 4 va 5
D) faqat 3 va 5

5. Reaksiyada uch element darajalarining o'zgarishi jadvalda berilgan:

Element	S	Mn	N
boshlang'ich	-2	+7	+3
oxirgi	0	+2	+5

Qaysi element(lar) QAYTARUVCHI bo'lgan?

A) faqat Mn
B) S va N
C) faqat S
D) Mn va N

6. $\text{KNO}_2 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \rightarrow \text{KNO}_3 + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$ reaksiyasida 0,5 mol oksidlovchi ishtirok etgan bo'lsa, sarflangan qaytaruvchining miqdorini (mol) toping.

A) 0,75 B) 2,5 C) 0,2 D) 1,25

7. Yuqoridagi (6-savoldagi) reaksiyada 1 mol qaytaruvchi ishtirok etgan bo'lsa, sarflangan oksidlovchining massasini (g) aniqlang. ($M(\text{KMnO}_4)=158$)

A) 63,2 B) 158 C) 31,6 D) 395

8. Quyidagi zarrachalardan qaysi biri FAQAT qaytaruvchi bo'la oladi?

A) SO_4^{2-} (S^{+6})
B) SO_3^{2-} (S^{+4})
C) S^{2-}
D) S^0

9. $\text{Cl}_2 + 2\text{KOH}(\text{sovuq}) \rightarrow \text{KCl} + \text{KClO} + \text{H}_2\text{O}$ reaksiyasi qaysi turga mansub?

A) molekulyararo OQR
B) ichki molekulyar OQR
C) disproporsiya (o'z-o'zidan oksidlanish-qaytarilish)
D) almashinish reaksiyasi

10. 0,2 mol alyuminiy to'liq oksidlanib Al^{3+} ga o'tganda nechta elektron beradi?

A) $0,2 \cdot N_A$ B) $1,8 \cdot N_A$ C) $3 \cdot N_A$ D) $0,6 \cdot N_A$

11. 2 mol noma'lum alkan kislorodda to'liq yondirildi. Bu jarayonda oksidlovchi qaytaruvchidan $76 \cdot N_A$ ta elektron olgan bo'lsa, alkanni aniqlang.

A) pentan B) geksan C) heptan D) oktan

12. $\text{FeCl}_3 + \text{H}_2\text{S} \rightarrow \dots$ reaksiyasini davom ettiring va barcha koeffitsiyentlar yig'indisini aniqlang.

A) 6 B) 8 C) 5 D) 11

13. $\text{K}_2\text{Cr}_2\text{O}_7 + \text{FeSO}_4 + \text{H}_2\text{SO}_4 \rightarrow \text{K}_2\text{SO}_4 + \text{Cr}_2(\text{SO}_4)_3 + \text{Fe}_2(\text{SO}_4)_3 + \text{H}_2\text{O}$ reaksiyasida QAYTARUVCHI va OKSIDLOVCHI oldidagi koeffitsiyentlarni mos ravishda aniqlang.

A) 1 va 6 B) 2 va 3 C) 6 va 7 D) 6 va 1

14. Quyidagi o'zgarishlarning qaysilari OKSIDLANISH jarayoniga mansub?

1) $\text{S}^{-2} \rightarrow \text{S}^{+4}$; 2) $\text{N}^{+5} \rightarrow \text{N}^{+2}$; 3) $\text{Al}^0 \rightarrow \text{Al}^{+3}$; 4) $\text{Cu}^{+2} \rightarrow \text{Cu}^0$.

A) 2 va 4
B) 1 va 2
C) faqat 3
D) 1 va 3

15. $\text{Cu} + \text{HNO}_3(\text{suyul.}) \rightarrow \text{Cu}(\text{NO}_3)_2 + \text{NO} + \text{H}_2\text{O}$ reaksiyasi tenglashtirilganda barcha koeffitsiyentlar yig'indisini toping.

A) 20 B) 13 C) 11 D) 9

16. Galogenlar orasida ENG KUHLI oksidlovchi qaysi?

- A) Cl_2 B) Br_2 C) F_2 D) I_2

17. $\text{Zn} + \text{HNO}_3$ (juda suyul.) $\rightarrow \text{Zn}(\text{NO}_3)_2 + \text{NH}_4\text{NO}_3 + \text{H}_2\text{O}$ reaksiyasining elektron balansi jadvalda:

Yarim reaksiya	Ko'paytiruvchi
$\text{Zn}^0 - 2e \rightarrow \text{Zn}^{+2}$	?
$\text{N}^{+5} + 8e \rightarrow \text{N}^{-3}$	?

«?» o'rnidagi ko'paytiruvchilarni mos ravishda aniqlang.

- A) 4 va 1 B) 1 va 4 C) 2 va 8 D) 8 va 2

18. NH_4NO_3 tarkibidagi azot atomlarining oksidlanish darajalarini aniqlang.

- A) +3 va -5
B) ikkalasi ham 0
C) -3 va +5
D) -3 va +3

19. $2\text{KMnO}_4 + 16\text{HCl} \rightarrow 2\text{KCl} + 2\text{MnCl}_2 + 5\text{Cl}_2 + 8\text{H}_2\text{O}$ reaksiyasi bo'yicha 0,5 mol xlor olish uchun necha gramm KMnO_4 kerak? ($M=158$)

- A) 79 B) 31,6 C) 63,2 D) 15,8

20. $\text{NH}_4\text{NO}_3 \rightarrow (\text{t}) \text{N}_2\text{O} + 2\text{H}_2\text{O}$ parchalanishi qaysi turdagi OQR?

- A) molekulyararo
B) ichki molekulyar
C) disproporsiya
D) OQR emas

21. $\text{As}_2\text{S}_3 + \text{HNO}_3 \rightarrow \text{H}_3\text{AsO}_4 + \text{SO}_2 + \text{NO}_2 + \text{H}_2\text{O}$ reaksiyasida 0,3 mol SO_2 hosil bo'lgan bo'lsa, oksidlovchining massasini (g) toping. ($M(\text{HNO}_3)=63$)

- A) 126 B) 246 C) 138,6 D) 492

22. Quyidagi reaksiyalarning qaysilari DISPROPORSIYAGA misol bo'ladi?

- 1) $\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{HCl} + \text{HClO}$; 2) $3\text{NO}_2 + \text{H}_2\text{O} \rightarrow 2\text{HNO}_3 + \text{NO}$;
3) $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$; 4) $4\text{KClO}_3 \rightarrow 3\text{KClO}_4 + \text{KCl}$.

- A) 1 va 3
B) 1, 2 va 4
C) faqat 1
D) 2, 3 va 4

23. Teng mol nisbatda olingan alkan va vodoroddan iborat 0,4 mol aralashma yondirilganda 1 mol suv hosil bo'ldi. Alkanni aniqlang.

- A) propan B) etan C) metan D) butan

24. $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$ reaksiyasida QAYTARUVCHI vazifasini qaysi zarracha bajaradi?

- A) MnO_2
B) HCl tarkibidagi Cl^-
C) H^+
D) H_2O

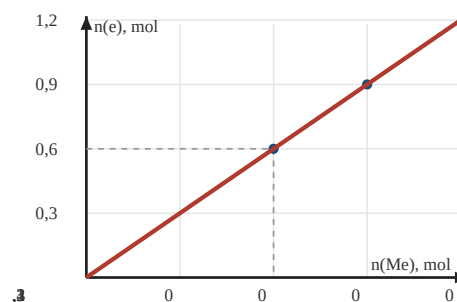
25. $2\text{AuCl}_3 + 3\text{H}_2\text{O}_2 + 6\text{KOH} \rightarrow 2\text{Au} + 3\text{O}_2 + 6\text{KCl} + 6\text{H}_2\text{O}$ reaksiyasida 13,44 l (n.sh.) gaz ajraldi. Qaytarilgan oltin atomlarining sonini aniqlang.

- A) $2,408 \cdot 10^{23}$
B) $6,02 \cdot 10^{23}$
C) $9,03 \cdot 10^{22}$
D) $1,204 \cdot 10^{23}$

26. $4\text{KClO}_3 \rightarrow (\text{t}) 3\text{KClO}_4 + \text{KCl}$ disproporsiya reaksiyasida 0,4 mol KClO_3 dan necha mol KClO_4 hosil bo'ladi?

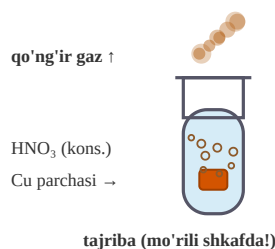
- A) 0,3 B) 0,4 C) 0,1 D) 1,2

27. Rasmda metall oksidlanayotganda bergan elektron mollari metall mol soniga bog'liq holda ko'rsatilgan. Grafikdan foydalanib, metallning valentligini va mos metallni aniqlang.



- A) 2; magniy
B) 1; natriy
C) 3; alyuminiy
D) 3; temir (faqat +3)

28. Rasmda mis parchasi konsentrlangan nitrat kislotaga tushirilganda qo'ng'ir gaz ajralishi ko'rsatilgan. Bu gaz va undagi azotning oksidlanish darajasi qaysi javobda to'g'ri?



- A) NO ; +2
B) N_2O ; +1
C) NO_2 ; +4
D) NH_3 ; -3

29. Permanganat-ionning turli muhitlardagi qaytarilish mahsulotlari jadvalda berilgan:

Muhit	Mahsulot	Olingan e soni
kislotali	Mn^{2+}	?
neytral	MnO_2	?
ishqoriy	MnO_4^{2-}	?

«?» o'rnidagi elektron sonlarini mos ravishda aniqlang.

- A) 3, 5 va 1
B) 5, 3 va 2
C) 7, 4 va 6
D) 5, 3 va 1

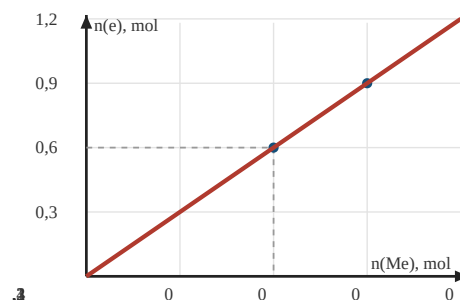
30. Vodorod peroksid (H_2O_2) tarkibidagi kislorodning oksidlanish darajasi qanday?

- A) -2 B) 0 C) +2 D) -1

31. 5,6 g temir xlorid kislota bilan to'liq reaksiyaga kirishdi ($Fe \rightarrow Fe^{+2}$). Bunda temir bergan elektronlar sonini aniqlang.

- A) $6,02 \cdot 10^{23}$
B) $2,408 \cdot 10^{23}$
C) $3,01 \cdot 10^{22}$
D) $1,204 \cdot 10^{23}$

32. 27-savoldagi grafikdan foydalaning: 0,3 mol metall to'liq oksidlanganda necha mol elektron beradi?



- A) 0,3 B) 0,6 C) 0,9 D) 1,2

2-QISM · GURUHLANGAN SAVOL (33-35 · A-F ro'yxatidan tanlang)

Laboratoriyada 0,3 mol mis bilan ikki tajriba o'tkazildi: 1-tajribada u SUYULTIRILGAN ($3Cu + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$), 2-tajribada KONSENTRLANGAN ($Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$) nitrat kislota to'liq eritildi. 33-35-savollarga A-F ro'yxatidan javob tanlang.

33. 1-tajribada necha mol kislota sarflangan?

34. 1-tajribada ajralgan gazning hajmi (l, n.sh.) qancha?

35. 2-tajribada ajralgan gazning hajmi (l, n.sh.) qancha?

Javoblar ro'yxati: A) 4,48 · B) 0,8 · C) 6,72 · D) 13,44 · E) 0,6 · F) 0,4

3-QISM · QISQA JAVOBLI SAVOLLAR (36-40)

36. $S^{+4} \rightarrow S^{-2}$ o'zgarishida oltingugurt atomi nechta elektron biriktiradi? Javob:

37. $2KMnO_4 + 16HCl \rightarrow 2KCl + 2MnCl_2 + 5Cl_2 + 8H_2O$. 0,1 mol $KMnO_4$ dan olinadigan xlorning hajmini (l, n.sh.) toping. Javob:

38. 6,2 g fosfor konsentrlangan nitrat kislota H_3PO_4 gacha oksidlandi ($P - 5e$). Ajralgan NO_2 ning hajmini (l, n.sh.) toping. ($M(P)=31$) Javob:

39. 5,4 g uch valentli metall to'liq oksidlanganda 0,6 mol elektron berdi. Metallning molyar massasini (g/mol) toping. Javob:

40. $FeS_2 + O_2 \rightarrow Fe_2O_3 + SO_2$ reaksiyasi tenglashtirilganda barcha koeffitsiyentlar yig'indisini toping. Javob:

4-QISM · YOZMA ISH (41–43 · har biri 25 ball)**41-topshiriq**

Laboratoriyada xlor olish uchun 15,8 g kaliy permanganat konsentrlangan xlorid kislota bilan reaksiyaga kiritildi: $\text{KMnO}_4 + \text{HCl} \rightarrow \text{KCl} + \text{MnCl}_2 + \text{Cl}_2 + \text{H}_2\text{O}$. Bandlar ketma-ket yechiladi. ($M(\text{KMnO}_4)=158$, $M(\text{HCl})=36,5$)

- Reaksiyani elektron balans usulida tenglashtiring. (5 ball · M4+A1)
- Hosil bo'ladigan xlorning miqdorini (mol) toping. (5 ball · M3+A2)
- Xlorning hajmini (l, n.sh.) hisoblang. (4 ball · M2+A2)
- Sarflangan kislotaning massasini (g) toping. (6 ball · M3+A3)
- Bu reaksiyada HCl qanday IKKI vazifani bajaradi? Izohlang. (5 ball · M3+A2)

42-topshiriq

Sxemadagi X_1 va X_2 moddalarni aniqlab, quyidagilarni bajaring:

$\text{KMnO}_4 \xrightarrow{(+\text{HCl kons.})} X_1(\text{sariq-yashil gaz}) \xrightarrow{(+\text{KOH, sovuq})} X_2 \xrightarrow{(t^\circ)} \text{KClO}_3$

- X_1 va X_2 ni aniqlab, 1- va 2-reaksiya tenglamalarini tenglashtirilgan holda yozing. (13 ball · M13)
- 3-reaksiya ($X_2 \rightarrow \text{KClO}_3$) tenglamasini tenglashtirilgan holda yozing. (9 ball · M9)
- Zanjirdagi qaysi reaksiyalar disproporsiyaga mansubligini ko'rsating. (3 ball · M3)

43-topshiriq

X, Y, Z metallarining xlorid kislota va mis (II) sulfat eritmasi bilan o'zaro ta'siri jadvalda berilgan («+» — reaksiya boradi, «-» — bormaydi):

Metall	HCl	CuSO_4 eritmasi
X	+	+
Y	-	+
Z	-	-

Bandlar ketma-ket yechiladi.

- X, Y, Z metallarini aktivlik qatorida joylashish tartibi bo'yicha yozing va asoslang. (4 ball · M4)
- $X = \text{Zn}$ bo'lsa, 13 g rux mo'l CuSO_4 eritmasidan necha gramm misni siqib chiqaradi? (7 ball · M4+A3)
- Shu jarayonda nechta elektron almashinadi (N_a birligida)? (6 ball · M3+A3)
- Nega Y metall kislota bilan reaksiyaga kirishmaydi-yu, CuSO_4 bilan kirishadi? Izohlang. (8 ball · M4+A4)

B-variant · Javoblar kaliti va yechimlar

Nº	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Javob	A	B	D	A	B	D	A	C	C	D	B	B	D	D	A	C

Nº	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
Javob	A	C	B	B	C	B	A	B	A	A	C	C	D	D	D	C

33-35: 33 → B, 34 → A, 35 → D · **36-40:** 36 → 6, 37 → 5,6, 38 → 22,4, 39 → 27, 40 → 25

1. (A) Balans: $\text{K}_2\text{Cr}_2\text{O}_7 + 3\text{H}_2\text{S} + 4\text{H}_2\text{SO}_4 \rightarrow \text{Cr}_2(\text{SO}_4)_3 + 3\text{S} + \text{K}_2\text{SO}_4 + 7\text{H}_2\text{O}$. O'ng: $1+3+1+7 = 12$. ($\text{Cr}^{+6}+3e \times 2$; $\text{S}^{-2}-2e \times 3$.)

2. (B) $2\text{KMnO}_4 + 16\text{HCl} \rightarrow 2\text{KCl} + 2\text{MnCl}_2 + 5\text{Cl}_2 + 8\text{H}_2\text{O}$. Jami: $2+16+2+2+5+8 = 35$. ($\text{Mn}^{+7}+5e \times 2$; $2\text{Cl}^{-}-2e \times 5$.)

3. (D) $2(+1) + 2x + 7(-2) = 0 \rightarrow 2x = 12 \rightarrow x = +6$.

4. (A) Elektroliz, yonish, korroziya — darajalar o'zgaradi (OQR); gidroliz va neytrallanish — almashinish.

5. (B) Qaytaruvchi — e beruvchi (daraja ortadi): $\text{S}(-2 \rightarrow 0)$ va $\text{N}(+3 \rightarrow +5)$. $\text{Mn}(+7 \rightarrow +2)$ — oksidlovchi.

6. (D) Balans: $5\text{KNO}_2 + 2\text{KMnO}_4 + 3\text{H}_2\text{SO}_4 \rightarrow \dots$ ($\text{N}: -2e \times 5$; $\text{Mn}: +5e \times 2$). $0,5 \cdot 5/2 = 1,25$ mol KNO_2 .

7. (A) $\text{KMnO}_4 = 1 \cdot 2/5 = 0,4$ mol $\rightarrow 0,4 \cdot 158 = 63,2$ g.

8. (C) S^{2-} — oltingugurtning eng past darajasi: faqat e berishi (oksidlanishi) mumkin.

9. (C) Cl^0 ning bir qismi -1 ga (qaytarildi), bir qismi $+1$ ga (oksidlandi) o'tdi — disproporsiya.

10. (D) $n(e) = 0,2 \cdot 3 = 0,6$ mol $\rightarrow 0,6 \cdot N_A$ ta elektron.

11. (B) $\text{C}_n\text{H}_{2n+2}$ yonishida 1 mol alkan $(6n+2)$ mol e beradi. $2(6n+2) = 76 \rightarrow n = 6$ → geksan.

12. (B) $2\text{FeCl}_3 + \text{H}_2\text{S} \rightarrow 2\text{FeCl}_2 + \text{S} + 2\text{HCl}$ ($\text{Fe}^{+3}+1e \times 2$; $\text{S}^{-2}-2e \times 1$). Jami: $2+1+2+1+2 = 8$.

13. (D) $\text{Fe}^{+2}-1e \times 6$; $\text{Cr}_2^{+6}+6e \times 1 \rightarrow \text{K}_2\text{Cr}_2\text{O}_7 + 6\text{FeSO}_4 + 7\text{H}_2\text{SO}_4 \rightarrow \dots$ Qaytaruvchi 6, oksidlovchi 1.

14. (D) Oksidlanish — daraja ortishi: 1 ($-2 \rightarrow +4$) va 3 ($0 \rightarrow +3$). 2 va 4 — qaytarilish.

15. (A) $3\text{Cu} + 8\text{HNO}_3 \rightarrow 3\text{Cu}(\text{NO}_3)_2 + 2\text{NO} + 4\text{H}_2\text{O}$ ($\text{Cu}-2e \times 3$; $\text{N}^{+5}+3e \times 2$). Jami: $3+8+3+2+4 = 20$.

16. (C) Ftor — eng elektromanfiy element: elektronni eng kuchli tortadi, faqat oksidlovchi bo'ladi.

17. (A) $\text{EKUK}(2,8) = 8 \rightarrow \text{Zn}$ uchun $8/2 = 4$, N uchun $8/8 = 1$. ($4\text{Zn} + 10\text{HNO}_3 \rightarrow 4\text{Zn}(\text{NO}_3)_2 + \text{NH}_4\text{NO}_3 + 3\text{H}_2\text{O}$.)

18. (C) NH_4^+ da $\text{N} = -3$; NO_3^- da $\text{N} = +5$ — bitta tuzda ikki xil daraja.

19. (B) $n(\text{KMnO}_4) = 0,5 \cdot 2/5 = 0,2$ mol $\rightarrow 0,2 \cdot 158 = 31,6$ g.

20. (B) Oksidlovchi (N^{+5}) va qaytaruvchi (N^{-3}) bitta molekula ichida — ichki molekulyar OQR.

21. (C) As_2S_3 22 e beradi ($\text{As}: 2e \times 2$, $\text{S}: 6e \times 3$) $\rightarrow \text{As}_2\text{S}_3 + 22\text{HNO}_3 \rightarrow 2\text{H}_3\text{AsO}_4 + 3\text{SO}_2 + 22\text{NO}_2 + 8\text{H}_2\text{O}$. $\text{HNO}_3 = 0,3 \cdot 22/3 = 2,2$ mol $\rightarrow 138,6$ g.

22. (B) 1: $\text{Cl}^0 \rightarrow -1/+1$; 2: $\text{N}^{+4} \rightarrow +5/+2$; 4: $\text{Cl}^{+5} \rightarrow +7/-1$ — bir element ikkiga ajraladi. 3 — oddiy o'rin olish.

23. (A) $0,2$ mol $\text{C}_n\text{H}_{2n+2} + 0,2$ mol H_2 : suv = $0,2(n+1) + 0,2 = 1 \rightarrow n = 3 \rightarrow$ propan.

24. (B) $\text{Cl}(-1) \rightarrow \text{Cl}_2(0)$: e berdi — qaytaruvchi. $\text{Mn}^{+4} \rightarrow \text{Mn}^{+2}$: e oldi — oksidlovchi.

25. (A) $\text{O}_2 = 0,6$ mol $\rightarrow \text{Au} = 0,6 \cdot 2/3 = 0,4$ mol $\rightarrow 0,4 \cdot 6,02 \cdot 10^{23} = 2,408 \cdot 10^{23}$ ta atom.

26. (A) Nisbat $4:3 \rightarrow 0,4 \cdot 3/4 = 0,3$ mol KClO_4 .

27. (C) Grafikdan: $0,2$ mol metall $\rightarrow 0,6$ mol e $\rightarrow n = 3$. Doimiy $+3$ valentli yengil metall — Al.

28. (C) Kons. HNO_3 misni oksidlaganda qo'ng'ir NO_2 (N^{+4}) ajraladi: $\text{Cu} + 4\text{HNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + 2\text{NO}_2 + 2\text{H}_2\text{O}$.

29. (D) $\text{Mn}^{+7} \rightarrow \text{Mn}^{+2}$ ($5e$); $\rightarrow \text{Mn}^{+4}$ ($3e$); $\rightarrow \text{Mn}^{+6}$ ($1e$).

30. (D) Peroksid ko'prigi $-\text{O}-\text{O}-$: har bir O ning darajasi -1 . Shu sababli H_2O_2 ham oksidlovchi, ham qaytaruvchi.

31. (D) $n(\text{Fe}) = 0,1$ mol $\rightarrow e = 0,2$ mol $\rightarrow 0,2 \cdot 6,02 \cdot 10^{23} = 1,204 \cdot 10^{23}$ ta.

32. (C) Grafik nisbati $3 \rightarrow 0,3 \cdot 3 = 0,9$ mol elektron.

33-35. 33: $\text{HNO}_3 = 0,3 \cdot 8/3 = 0,8$ mol (B). 34: $\text{NO} = 0,2$ mol $\rightarrow 4,48$ l (A). 35: $\text{NO}_2 = 0,6$ mol $\rightarrow 13,44$ l (D).

36. Daraja $+4$ dan -2 gacha pasaydi: $4 - (-2) = 6$ e.

37. $\text{Cl}_2 = 0,1 \cdot 5/2 = 0,25$ mol $\rightarrow 0,25 \cdot 22,4 = 5,6$ l.

38. $\text{P} = 0,2$ mol $\rightarrow e = 1$ mol $\rightarrow \text{NO}_2$ ($1e$) = 1 mol $\rightarrow 22,4$ l.

39. $n(\text{Me}) = 0,6/3 = 0,2$ mol $\rightarrow M = 5,4/0,2 = 27$ g/mol (Al).

40. $4\text{FeS}_2 + 11\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3 + 8\text{SO}_2$ ($\text{Fe}^{+2} \rightarrow \text{Fe}^{+3}$, $\text{S}^{-1} \rightarrow \text{S}^{+4}$: $11e \times 4$; $\text{O}_2 + 4e \times 11$). Jami: $4+11+2+8 = 25$.

41-topshiriq. a) Reaksiyani elektron balans usulida tenglashtiring. $\text{Mn}^{+7}+5e \times 2$; $2\text{Cl}^{-}-2e \times 5 \rightarrow 2\text{KMnO}_4 + 16\text{HCl} \rightarrow 2\text{KCl} + 2\text{MnCl}_2 + 5\text{Cl}_2 + 8\text{H}_2\text{O}$. **b)** Hosil bo'ladigan xlarning miqdorini (mol) toping. $n(\text{KMnO}_4) = 0,1$ mol $\rightarrow \text{Cl}_2 = 0,1 \cdot 5/2 = 0,25$ mol. **c)** Xlarning hajmini (l, n.sh.) hisoblang. $V = 0,25 \cdot 22,4 = 5,6$ l. **d)**

Sarflangan kislotaning massasini (g) toping. $\text{HCl} = 0,1 \cdot 16/2 = 0,8 \text{ mol} \rightarrow 29,2 \text{ g}$. **e) Bu reaksiyada HCl qanday IKKI vazifani bajaradi? Izohlang.** 16 HCl dan 10 tasi qaytaruvchi ($\text{Cl}^- \rightarrow \text{Cl}_2$), 6 tasi muhit — tuz hosil qiladi; (KCl, MnCl_2 tarkibiga kiradi)..

42-topshiriq. a) X_1 va X_2 ni aniqlab, 1- va 2-reaksiya tenglamalarini tenglashtirilgan holda yozing. $\text{X}_1 = \text{Cl}_2$, $\text{X}_2 = \text{KClO}$. $2\text{KMnO}_4 + 16\text{HCl} \rightarrow 2\text{KCl} + 2\text{MnCl}_2 + 5\text{Cl}_2 + 8\text{H}_2\text{O}$; $\text{Cl}_2 + 2\text{KOH} \rightarrow \text{KCl} + \text{KClO} + \text{H}_2\text{O}$. **b) 3-reaksiya ($\text{X}_2 \rightarrow \text{KClO}_3$) tenglamasini tenglashtirilgan holda yozing.** $3\text{KClO} \rightarrow (\text{t}) 2\text{KCl} + \text{KClO}_3$ ($\text{Cl}^{+1} \rightarrow -1$ va $+5$ — disproporsiya). **c) Zanjirdagi qaysi reaksiyalar disproporsiyaga mansubligini ko'rsating.** 2- va 3-reaksiyalar: $\text{Cl}^0 \rightarrow -1/+1$ va $\text{Cl}^{+1} \rightarrow -1/+5$..

43-topshiriq. a) X, Y, Z metallarini aktivlik qatorida joylashish tartibi bo'yicha yozing va asoslang. $\text{X} > \text{Y} > \text{Z}$: X vodoroddan oldin; Y — H bilan Cu orasida; Z — Cu dan keyin.. **b) X = Zn bo'lsa, 13 g rux mo'l CuSO_4 eritmasidan necha gramm misni siqib chiqaradi?** $n(\text{Zn}) = 0,2 \text{ mol} \rightarrow \text{Cu} = 0,2 \text{ mol} \rightarrow 12,8 \text{ g}$. **c) Shu jarayonda nechta elektron almashinadi (N_a birligida)?** $e = 0,2 \cdot 2 = 0,4 \text{ mol} \rightarrow 0,4 \cdot \text{N}_a$ ($\approx 2,408 \cdot 10^{23}$) ta. **d) Nega Y metall kislota bilan reaksiyaga kirishmaydi-yu, CuSO_4 bilan kirishadi? Izohlang.** Y aktivlik qatorida H dan keyin — H^+ ni qaytara olmaydi; ammo Cu dan oldin —; Cu^{2+} ni qaytara oladi (masalan, Bi, Sb kabi)..

Manba: Tongotarov OQR banki (2019-2021) arxetiplari — javoblar elektron balans bilan mustaqil qayta hisoblangan; MS spetsifikatsiyasi I.9